INTERNATIONAL CONFERENCE ON “INTEGRATING AI AND BIOMEDICAL ENGINEERING FOR SMARTER HEALTHCARE”

(NCABSH 2026-HYBRID MODE)

HOSTED BY

BHARATH INSTITUTE OF HIGHER EDUCATION AND RESEARCH

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TITLE

PERFORMANCE EVALUATION AND OPTIMIZATION OF DUAL-GATE CORBON NANO TUBE FILED EFFECT TRANSISTOR-BASED GAS SENSORS

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DEPARTMENT OF BIOMEDICAL ENGINEERING

Mode: **Online Mode**

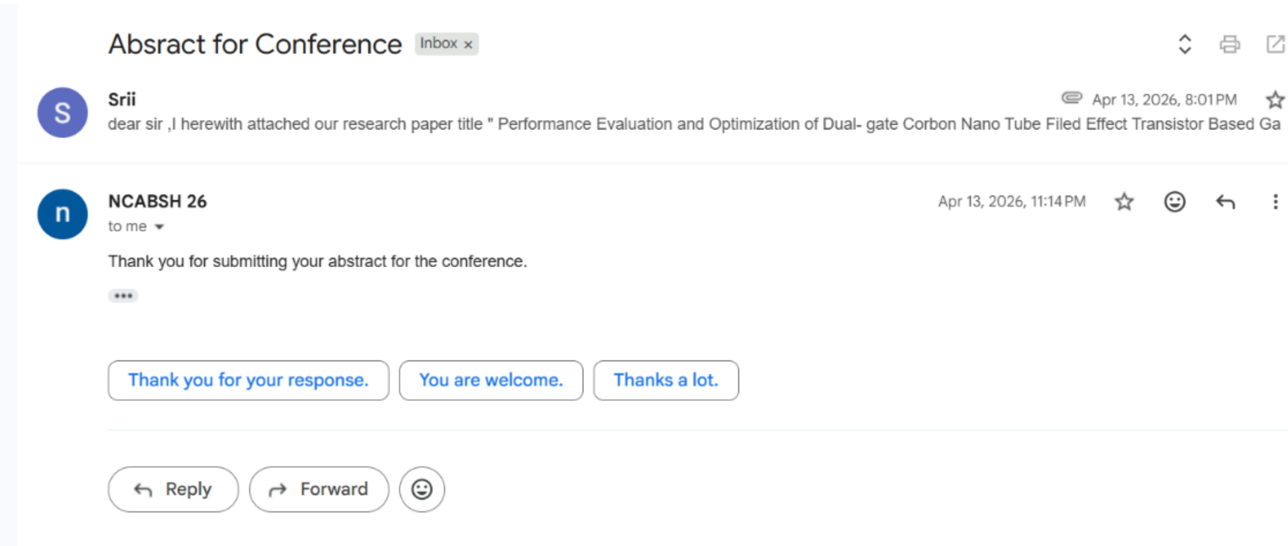
Session & Time: XII – 1.30pm to 4.30 pm

ABSTRACT

Field Effect Transistors (FETs) have lived to see its significant technological improvement for various applications in recent years. Carbon nanotube (CNT) based FET have been found to be a promising component for next generation transistor technologies for CNTs high carrier mobility, device stability and mechanical flexibility. However the design of CNTFET is still not well established, especially with a view to achieve the best performance still protecting thermal stability. In this study the authors had analysed the dual gate device structure and operation of transistor in which carbon nanotubes act as active channel region. DG CNTFET with different device geometrics such as channel length, oxide thickness for its output and transfer characteristics have been extensively studied using Nano TCAD ViDES simulation software. This study has thrown new insight into the device performance characteristics of DG CNTFET which can be scaled down up to minimum channel length without short channel effects.

Keywords: IC-Scaling, CNT, Nano Transistors, Flexible Electronics, Network Transistor, CNT-FET

NCABSH -2026 REGISTRATION CONFIRMATION


Absract for Conference Inbox ×

Srii
Apr 13, 2026, 8:01 PM
dear sir ,I herewith attached our research paper title " Performance Evaluation and Optimization of Dual- gate Corbon Nano Tube Filed Effect Transistor Based Ga

NCABSH 26
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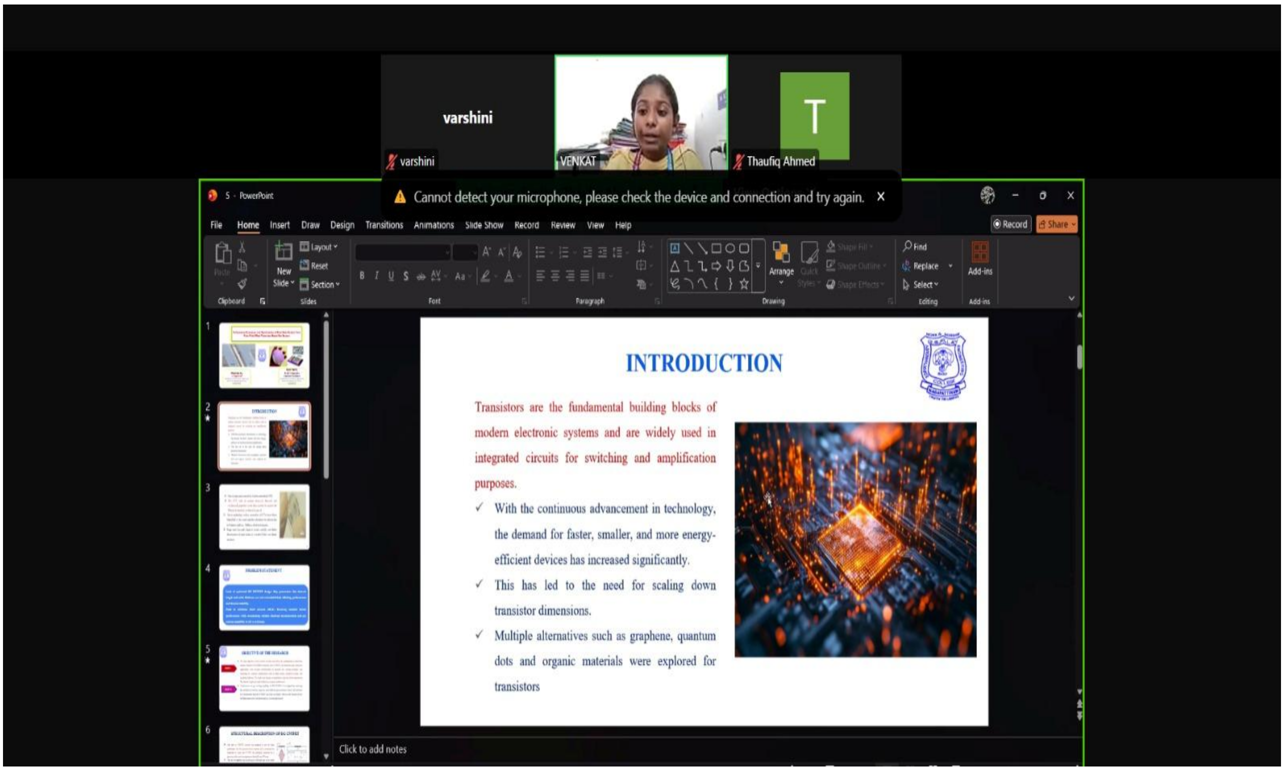
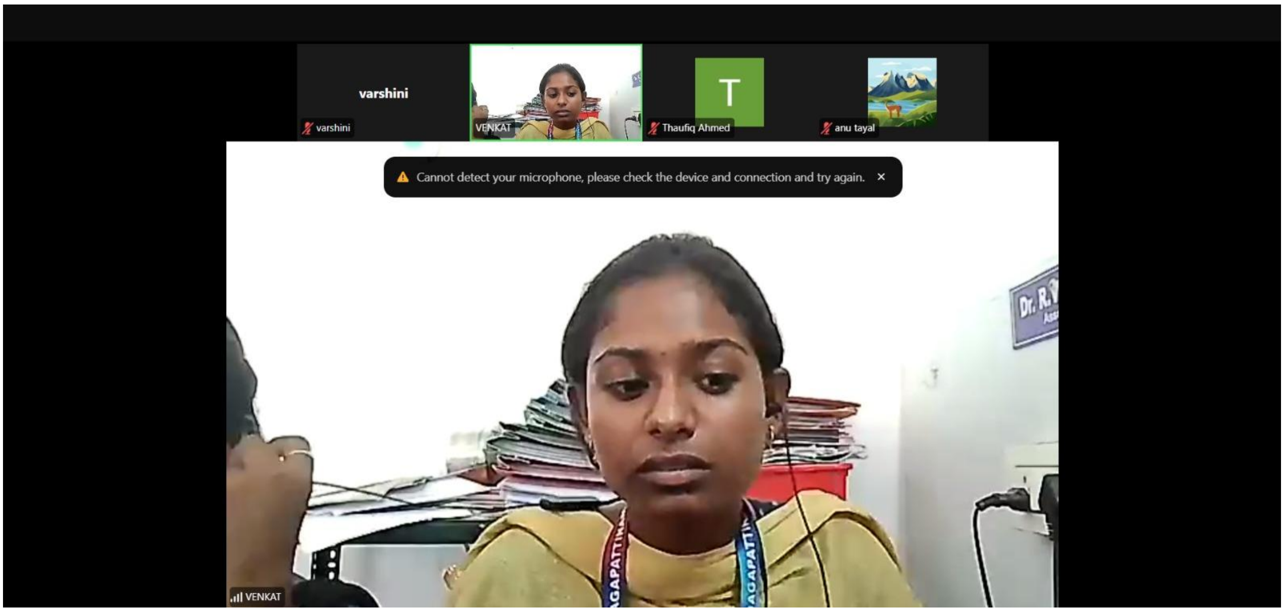
Thank you for submitting your abstract for the conference.

Thank you for your response. You are welcome. Thanks a lot.

Reply Forward

PRESENTATION @ NCABSH 2026





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Dr. R. Vasuki Dean- Admission Prof & HOD - BME	Dr. Jeyanthi Rebecca Dean- School of Bioengineering	Dr. M. Sundararaj Dean Academics

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